Morphological Chart – Heatwave Monitoring & Heatmap System

Problem Statement

How might we identify heat-stressed areas in a city and provide timely information to citizens and authorities?

Main Functions and Possible Alternatives

Function	Alternative 1	Alternative 2	Alternative 3	Alternative 4
Temperature Data Collection	IoT Sensors	Weather Stations	Satellite Data	Crowd sourced Data
Location Tracking	GPS	GIS Coordinates	Mobile Location	Fixed Zones
Heat Calculation	Threshold Rules	Heat Index	ML Prediction	Spatial Analysis
Heatmap Generation	GIS Heatmap	Web Map	Raster Map	Interactive Map
Data Processing	Cloud	Edge	Local Server	Hybrid
Prediction	Statistical Model	ML	Deep Learning	No Prediction
Alert Generation	SMS	Push Notification	Email	Voice Alert
User Interface	Mobile App	Web Dashboard	GIS Dashboard	Chatbot
Target User	Citizen	Municipality	Health Department	Disaster Management
Visualization	Color Heatmap	Risk Zones	Charts	3D Map
Historical Analysis	Daily	Weekly	Monthly	Seasonal
Recommendation	Stay Indoors	Cooling Center	Route Suggestion	Water Reminder

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1. How the Chart Generates Alternatives

We **don't simply select one option from each row randomly**.

We create meaningful combinations.

Alternative A – IoT Smart Heatmap Developer 1

```
IoT Sensors
      ↓
GPS
      ↓
Heat Index
      ↓
Cloud Processing
      ↓
GIS Heatmap
      ↓
Web Dashboard
      ↓
Municipality
```

Concept

A city installs temperature and humidity sensors at different locations. The system calculates heat index and displays the results as an interactive heatmap.

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2. Alternative B – Satellite-Based Heat Monitoring

```
Satellite Data
      ↓
GIS Coordinates
      ↓
Spatial Analysis
      ↓
Cloud Processing
      ↓
Raster Heatmap
      ↓
Web GIS
      ↓
Disaster Management
```

Concept

Satellite imagery is analyzed to identify areas experiencing higher surface temperatures.

Advantage: Large geographical coverage.

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3. Alternative C – Citizen-Based Heatmap

```
Mobile Location
       +
Crowdsourced Temperature
       ↓
ML Processing
       ↓
Interactive Heatmap
       ↓
Mobile Application
       ↓
Citizen
```

Citizens contribute local observations through a mobile application.

The system combines observations to create a **community-generated heatmap**.

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4. Alternative D – Predictive Heatwave System

This is a more advanced alternative.

```
Weather Stations
       +
Satellite Data
       +
Historical Data
       +
IoT Sensors
       ↓
Machine Learning
       ↓
Heatwave Prediction
       ↓
Risk Heatmap
       ↓
Early Warning
       ↓
Citizens + Authorities
```

Instead of showing only:

"Where is it hot now?"

the system answers:

"Which areas are likely to experience dangerous heat tomorrow?"

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5. Alternative E – Smart Heatwave + Safe Route

This is an interesting **innovative design alternative**.

Suppose a person wants to travel from:

A → B

The system considers:

- Heatmap
- Temperature
- Shade
- Time of day
- Green areas
- Cooling centers

and recommends:

"Take Route B. It is 7 minutes longer but has lower heat exposure."

Architecture

```
        Heatmap
           +
     Road Network
           +
    Weather Data
           ↓
      Route Engine
           ↓
   Heat-Safe Route
           ↓
       Citizen
```

This changes the system from a **visualization tool** into a **decision-support system**.

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6. Alternative F – Heatwave Health Risk System

Add health-related information.

```
Temperature
Humidity
Heat Index
```

```
      ↓
Risk Model
      +
Population Information
      ↓
Heat Risk Map
      ↓
High-Risk Zones
      ↓
Targeted Alerts
```

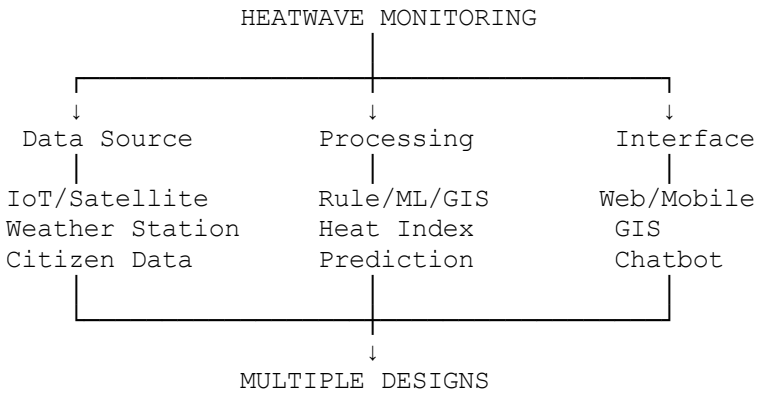
For example:

🔴 **High Risk Zone**
🟠 **Moderate Risk**
🟢 **Low Risk**

Authorities can prioritize interventions in high-risk areas.

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7. How the Design Space Works



Students can then see that **one problem can lead to many designs**.

Classroom Activity – "Build Your Heatwave System"

Divide the class into groups.

Give each group the same morphological chart.

Challenge

"Create one unique heatwave monitoring system by selecting suitable alternatives from the chart."

Group 1

IoT + Heat Index + Cloud + GIS + Mobile

Group 2

Satellite + ML + Cloud + Raster Map + Web

Group 3

Citizen Data + ML + Edge + Interactive Map + Mobile

Group 4

IoT + Satellite + Deep Learning + Cloud + GIS Dashboard

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9. Add a Design Thinking Twist

After each group creates its system, ask:

Question 1 – User

Who is your primary user?

Citizen / Municipality / Disaster Management / Health Department

Question 2 – Pain Point

What problem does your system solve?

Question 3 – Technology

Why did you select these technologies?

Question 4 – Feasibility

Can this realistically be implemented?

Question 5 – Innovation

What makes your design different from existing systems?

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10. Final Evaluation Matrix

Have students compare their alternatives:

Criteria	Alternative A	B	C	D
User Value	5	4	5	5
Feasibility	5	4	3	3
Cost	4	3	5	3
Scalability	4	5	3	5
Innovation	3	4	5	5
Total	**21**	**20**	**21**	**21**

Then ask:

"Which alternative would you prototype first, and why?"

This connects **Generating Design Alternatives → Evaluation → Prototyping**.

The key teaching message

A morphological chart does not give you the solution. It gives you a structured design space from which multiple possible solutions can be constructed.

For your **heatwave monitoring presentation**, this can lead nicely into **brainstorming → morphological chart → design alternatives → feasibility evaluation → prototype**.